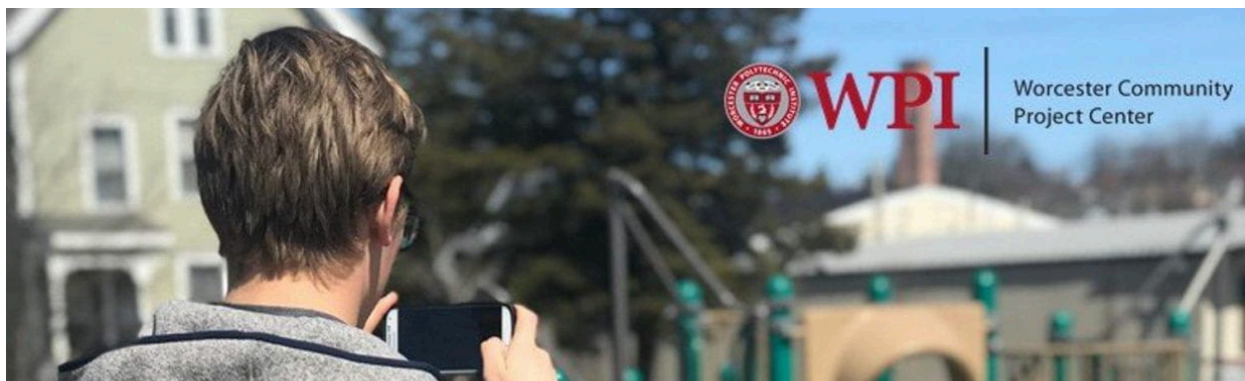




# ***CELEBRATING THE IMPACT OF THE WORCESTER COMMUNITY PROJECT CENTER***



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## **Celebrating the 25th Year of the Worcester Community Project Center**

An Interactive Qualifying Project Report  
submitted to the Faculty of  
WORCESTER POLYTECHNIC INSTITUTE  
in partial fulfillment of the requirements for the  
degree of Bachelor of Science

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This report represents the work of WPI undergraduate students submitted to the faculty as evidence of a degree requirement. WPI routinely publishes these reports on its website without editorial or peer review. For more information about the projects program at WPI, see <http://www.wpi.edu/Academics/Projects>.

# ***Abstract***

This project addresses and documents the impact of the Worcester Community Project (WCPC) over the last 25 years in the Greater Worcester Area. Since its establishment, the project center has been partnering with local organizations and non-profits to address the local challenges. Thus, our project researched the different and various projects that have been done by student researchers and conducted structured interviews with community partners and sponsors. We analyzed the ripple effects and inter-organizational connections generated by these projects. Our study identified patterns of sustained collaborations and long-term community impact. And to preserve and communicate these findings, our team developed a comprehensive digital website documenting project histories and measurable outcomes in order to emphasize the effect these had and make it accessible and understandable to everyone.

# ***Executive Summary***

Over the past 25 years, the Worcester Community Project Center (WCPC) has facilitated an important number of student research projects addressing different social and organizational issues in the Greater Worcester Area. The WCPC has also been the bridge between WPI students and organizations, where this partnership is formed to solve pragmatic solutions to pressing community challenges. Thus, the Center uses an approach named Community-Engaged Learning, a high-impact learning technique that combines meaningful community engagement with skills learned from academic curriculum to promote civic involvement and critical thinking (Elmes & Loiacono, 2009). Students are not just completing assignments—they're collaborating with organizations across Worcester to solve problems that actually exist outside the classroom, on issues that are real, urgent, and tied directly to Worcester's well-being. And what students create, whether it's a model, a dashboard, a program, or a tool, gets used by real people trying to serve real communities. This connection between learning and impact is the heart of WCPC's work.

## **Motivation**

More than 260 projects have been completed, but despite the multitude of these projects and the impact these projects have had, no documentation of the efforts of the full scope has been done. Hence, this project aimed to fill that gap by researching the breadth of the WCPC's initiatives, by identifying their effects on local organizations, and finding the networks of collaborations created over time. The research combined both coding analysis of the past project reports and interviews of sponsors and community partners. Assessing both qualitative and quantitative data and outcomes. We have conducted 11 semi-structured interviews across 10 different organizations assessing the impact of 11 projects. Key findings revealed that WCPC projects have significantly strengthened nonprofit operations by helping with a lack of time and bandwidth. Student researchers have helped organizations develop tools,<sup>1</sup> models,<sup>2</sup> databases,<sup>3</sup> and storytelling resources that helped organizations better organize information and communicate their work more effectively.<sup>4</sup> These projects didn't just benefit the education of the student researchers but also provided real value to organizations and the community.

## **Our Approach**

To properly document the stories and impacts that WCPC projects have on the Greater Worcester Area for its 25th anniversary, we, as a team, analyzed both long-term patterns across projects and firsthand perspectives from sponsors. This process transformed 25 years of individual projects into a structured dataset, allowing us to identify patterns across organizations and recurring community needs. By speaking directly with partners working within these recurring areas, we were able to better understand the real-world impact of WCPC projects beyond what the dataset alone could show. To get feedback on promotional materials, we gathered feedback on our website from sponsors and our advisor to ensure that the Center's 25-year history, partnerships, and impact were represented clearly, accurately, and in a way that is accessible to both community members and future sponsors.

### **Finding 1: Sustained Impact Across Three Core Themes**

Through our content analysis, we learned that through the 264 projects completed, WCPC has supported 168 organizations across 13 different themes we derived. The table below (Figure E.1) helps us visualize which issues were most common among organizations. Three themes clearly stood out:

1. Environment & Sustainability (50 projects)
2. Education & Learning (38 projects)
3. Community Development (34 projects)

**Figure E.1**

*Derived Organization Themes*

| Themes                                   | Description                                                                   | Count |
|------------------------------------------|-------------------------------------------------------------------------------|-------|
| Accessibility                            | Assistive technology, accessibility services                                  | 13    |
| City Planning                            | Zoning, regional planning                                                     | 11    |
| Health & Wellness                        | Mental health, caregiving, public health, well-being                          | 18    |
| Education & Learning                     | Academic Institutions, STEAM, literacy, curriculum                            | 38    |
| Environment & Sustainability             | Climate, renewable energy, conservation, water, waste                         | 50    |
| Social Equity & Inclusion                | Refugees, immigrants, gender equity, LGBTQ+, racial equity                    | 12    |
| Housing & Infrastructure                 | affordable housing, shelter, infrastructure                                   | 16    |
| Organizational Development & Non-profits | Marketing, nonprofit management, entrepreneurship, organization effectiveness | 16    |
| Food Insecurity                          | Food banks, food hubs                                                         | 12    |
| Economic Growth                          | Local Economy                                                                 | 9     |
| Culture & Arts                           | Museums, cultural exhibits, art, historic preservation                        | 26    |
| Community Development                    | Civic engagement, volunteering, neighborhood engagement                       | 34    |
| Urban Planning & Mobility                | Transportation, public spaces, parks, urban planning                          | 9     |

*Note.* These themes were derived based on the results of our content analysis

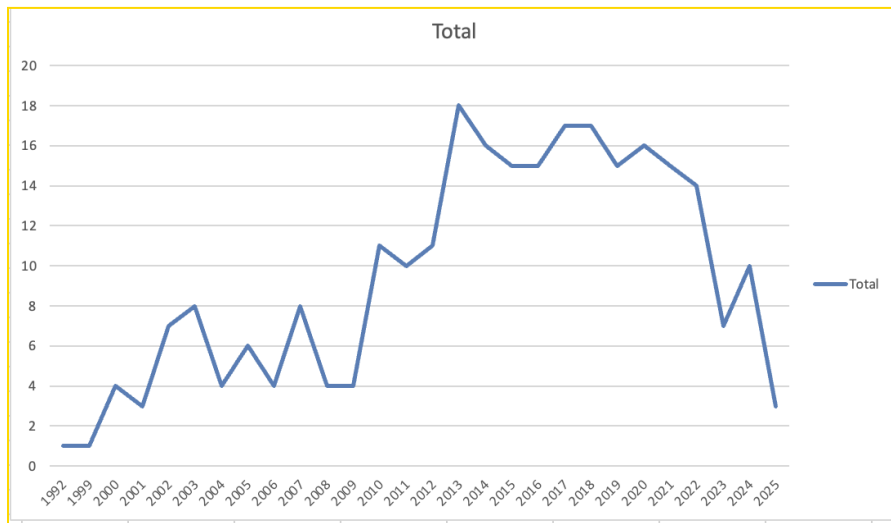
This wasn't random; it reflects what Worcester needs. These themes also helped us understand what these organizations were looking for. Education-focused partners needed tools to support learning. Community development organizations often need help with outreach, accessibility, or strengthening neighborhood services. Many projects focused on program development, like building systems, models, and tools that strengthen organizational capacity, and other projects helped shape climate-planning tools and water-quality resources, making complex data usable for the communities that need it. So this finding didn't just show us common themes; it helped us understand the landscape of community needs and guided which partners we reached out to next.

We also examined project activity over time (Figure E.2) and found that although WCPC projects steadily increased for nearly two decades, activity declined after 2020. An interview with the Director of the Worcester Community Project Center revealed that the introduction of

the \$5,000 Global IQP Scholarship shifted student interest toward international projects. While COVID-19 temporarily paused that shift, project numbers declined once travel resumed. This finding highlights the challenges the WCPC faces in sustaining its impact on the Worcester community.

**Figure E.2**

*Project Count Per Year Graph*



*Note.* The vertical axis refers to project count, while the horizontal axis refers to project year.

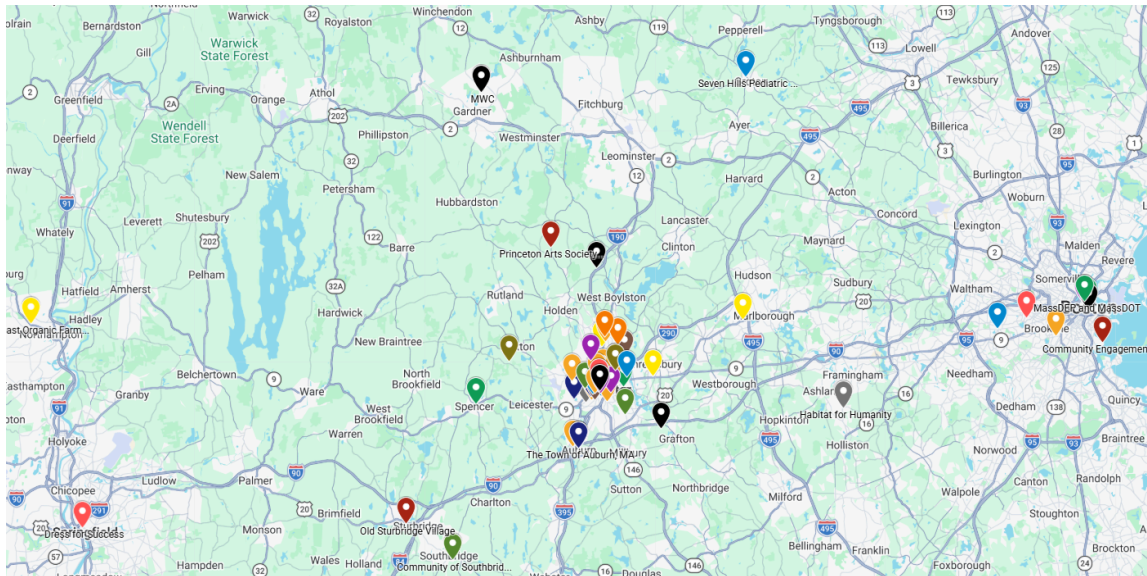
While this information and trend represent a clear challenge and a potential threat to the Center's continuous impact through the upcoming years, the university has recently announced changes to the global scholarship and the recipients' regulations, which might affect this graph in the future and reverse this trend.

From here, we wanted to show the landscape of the community presence these organizations bring to Massachusetts and specifically Worcester. By geographically mapping all 168 organizations that WCPC has partnered with (Map E.1), we were able to see that many of these organizations are located in downtown Worcester, with a few scattered in Boston and other cities.



## Map E.1

### *Geographic Organization Map*



*Note.* A geographic map of organizations that worked with WCPC was created in Google Maps. Link to map [HERE](#)

## **Finding 2: WCPC Creates Long-Term Relationships and Real-World Impact**

Worcester Community Project Center (WCPC) creates long-term relationships with real-world impact. Organizations, such as Habitat for Humanity, the Worcester Art Museum, and the Seven Hills Foundation, collectively make up about 12% of all collaborations with the WCPC since 2000. These repeated collaborations do more than demonstrate consistency; they reflect trust, credibility, and sustained support. When organizations choose to return to WCPC for multiple projects, it signals that prior collaborations were not only successful but meaningful enough to continue investing time, resources, and ideas into the partnership. This continuity across years and thematic areas suggests that WCPC is not functioning as a one-time resource for organizations, but as a reliable, embedded partner working to solve issues within the Worcester Community.

An example of the real-world impact that WCPC helped provide is an IQP team that created a calculation model with Habitat for Humanity ReStore. The outcome helped measure environmental impact and strengthen funding narratives. The tool was later adopted by ReStores regionally, nationally, and internationally, becoming widely used for its simplicity and impact.



Interviews revealed that many organizations struggled with communication challenges and data organization. For instance, the Broad Meadow Brook Trail Development project demonstrates how partnerships can extend beyond a single organization. Through collaboration with Mass Audubon, students worked to create a physical and educational connection between the Broad Meadow Brook Wildlife Sanctuary and the future Blackstone Valley Visitor Center. This project did more than simply map and blaze a trail; it connected conservation efforts, public education, and community access to nature.

### **Finding 3: Filling Capacity Gaps Through Student Expertise**

Organizations emphasized that WCPC projects help address staffing shortages due to time constraints and funding limitations. Many nonprofits are focused on direct service and lack the time or technical capacity to pursue projects such as archival digitization, environmental systems, or technology development.

For example, Abby's House needed to digitize archives for its 50th anniversary, but could not divert staff from serving women in need. IQP students dedicated focused time and technical skills to complete this work, filling a gap the organization could not address independently. Similarly, Habitat for Humanity's translation tool expanded engagement and goodwill within the community. As one sponsor noted, the tool *"changed the relationship between the shoppers, the donors, and our organization."* Another example is a centralized public event calendar that allowed many universities to post their public events, which made it easier for the community to be aware of these events. This event calendar, although short-lived, improved engagement between universities and the community.

Across five interviews, sponsors consistently described how student-created tools enabled them to serve the community despite limited resources. Students applied their classroom knowledge to build things that lasted. In this way, WCPC expands organizational capacity and strengthens community outreach.

### **Recommendations**

Based on these findings, we propose four recommendations to strengthen how WCPC documents, communicates, and sustains its impact:

**Revamp the project archive system:** Since the WCPC website only has projects from 2012 and later, we recommend updating the WCPC website to reflect all completed projects, linking

entries to Digital WPI, and applying standardized themes and project-type tags that we have created to improve accessibility and clarity.

**Publish a Public Partners Page:** Create a searchable directory highlighting partner organizations, years of collaboration, and engagement levels to make WCPC's involvement more visible.

**Create Space for Partners to Connect Through an Annual WCPC Community Gathering:** Host an annual networking event, allowing organizations to interact and connect with community members. This event would provide an opportunity to showcase the project deliverables of past projects and give feedback on the project deliverables. This will highlight WPI's role in local capacity-building and WCPC's sustained community development.

**Embed Community Voice Through Formalized Feedback Processes:** Implement structured post-project interviews or surveys to gather sponsor reflections on communication, engagement, and overall experience, strengthening long-term partnership quality.

To preserve and communicate these findings, a publicly accessible website that documents the history, legacy, and relationships of the WCPC projects was created. This website explains what WCPC is, what it does, and our motivation for the site. Then, it goes more in-depth into where and how WCPC makes an impact, how it connects and impacts the community of Worcester through partnerships, and what organizations say about the skills, innovations, and implementations that student researchers provide. This resource provides the base and foundation for future engagement, ensuring that the legacy of WCPC's work continues to benefit the Greater Worcester Area. Scan the QR code (Figure E.4) OR click on the link to explore our deliverable website: [WCPC IQP Website](#).

**Figure E.4**

*QR Code Link*



*Note.* QR code linking to the WCPC IQP Website

# ***Acknowledgements***

We would like to acknowledge everyone who helped guide us throughout our project:

**Professor Amanda Wittman**, our sponsor and an Assistant Teaching Professor in the Department of Global and Integrative Studies at WPI, provided thoughtful guidance, valuable feedback, and expertise in data-driven storytelling strategies that strengthened and shaped our project.

**Professor Courtney Kurlanska**, Assistant Professor of Teaching in the Department of Integrative & Global Studies, helped oversee and advise on the integrity of the project, as well as provide a great amount of feedback for our deliverable and full report.

**Professor Laura Roberts** is the director of the Worcester Community Project Center and the Co-director of the Montreal Project Center. Laura helped provide contact information for organizations to reach out to, and provided feedback for our final deliverable. She made sure we emphasized the importance of the WCPC, pushing for a major celebration for its 25th year.

And of course, those who were able to conduct interviews and focus groups with us! Thank you!

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Together, we worked through seven weeks to achieve this project!

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### **AI Assistance Statement:**

*ChatGPT (OpenAI) was used to assist with grammar, phrasing, and minor edits. All ideas, writing, and analysis are our own.*

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# Introduction

After doing something kind, we often feel a sense of accomplishment—especially when we can celebrate that moment with others. We may want to tell people about the good we’ve done, excited to share that feeling, yet those who weren’t involved might not understand why the action mattered or why it felt meaningful to us. The most effective way to convey the significance of a good deed, especially one that is complex or close to our heart, is to explain its value and impact clearly. The same is true for large-scale community work, where the meaning isn’t always visible unless we take the time to stop, reflect, and tell its story. The upcoming 25th anniversary of the Worcester Community Project Center (WCPC) offers an opportunity to reflect on a remarkable legacy: over 250 completed projects, more than 100 community partnerships, and countless stories of student learning, community collaboration, and organizational growth. How does a community project center move from a one-time initiative to becoming a lasting force for change? For the WCPC, this question is central to its mission of fostering long-term, meaningful partnerships between universities and the greater Worcester community.

Although the WCPC has facilitated numerous and diverse projects over the years, for example, identifying a low-cost translation tool for Humanity ReStore to increase accessibility, the full scope of its impact—the changed lives, the strengthened social networks, the lasting community improvements, and the legacy it leaves behind—is often unrecognized. While the WCPC tracks project deliverables and outcomes, it lacks a systematic framework for capturing the long-term impacts and stories of its work. Without a structured and intentional way to capture and communicate this influence, the WCPC risks overlooking some of its most significant achievements, leading to missed opportunities to deepen community engagement. This project addresses that critical gap to help garner the support needed for the WCPC to continue its work helping the community. Thus, our purpose as a team will be to develop a robust framework to assess the WCPC’s participatory impact on both sides of its partnerships: community members and organizations. We will move beyond basic metrics and highlight the rich human narratives that truly reflect change and can be transformed into a compelling story that resonates with communities, future WCPC teams, and prospective sponsors.

To do this, we first conducted research, as discussed in the background chapter—including high-impact learning—to clarify its benefits, values, and challenges before assessing its effects on organizations and community members. We explored impact

assessment, why it is important, and how projects can effectively communicate to a wide audience. Using this context, we examined the projects the WCPC has been involved in by conducting interviews and Ripple Effects Mapping to understand the impact from the perspective of project sponsors. We also developed a final website deliverable and conducted feedback interviews using an object–test–response approach to gather constructive insights on its effectiveness and usability. In the findings section, we discuss the visual representations we developed to support narrative storytelling, along with key insights we found shared by project sponsors. Lastly, we outline recommendations for the WCPC moving forward. Overall, we employed a mixed-methods strategy that incorporated interviews, and a focus group to better understand participant experiences, challenges, and successes. We then synthesized and communicated these insights through accessible formats, including visual representations and narrative storytelling, and developed a framework to evaluate the accessibility and clarity of our media products, ensuring they authentically reflected the WCPC's legacy and celebrated its achievements.

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# Background

This background chapter provides the contextual foundation for the project. It introduces the models of High-Impact Learning and Community-Engaged Learning, outlines their core principles and benefits for students and communities. Then moved to practices of impact assessment, a framework that describes how well a project serves the community as intended. Finally, it explores the field of strategic communication and promotion, focusing on how impact can be translated into compelling stories.

## High-Impact Learning

What would you say if I told you there is a practice that pushes students beyond the classroom and into the real world, where their actions meet the complex situations communities face every day? At the core of this is the practice of high-impact learning—experiential approaches that shape how students engage with real-world challenges across diverse settings and communities. High-Impact Learning (HIL) broadly describes practices that offer substantial academic advantages for participating students, especially those from populations historically underrepresented in higher education (College of Science and Mathematics, 2023). These approaches provide rich, transformative learning experiences that extend beyond traditional classroom activities, strengthening engagement, persistence, and overall learning. According to Reilly, Langley-Turnbaugh, and Neikirk (2023), high-impact learning enhances students' academic performance, writing skills, critical thinking, leadership, communication abilities, and personal development. As shown in Figure 1, Pusca and Northwood (2018) outline core characteristics of high-impact practices—dedicated time, exposure to new ideas, real-world application, and continuous assessment—that help explain their strong educational and community outcomes. HIL includes practices such as Project-Based Learning (PBL) and Community-Engaged Learning (CEL), which offer substantial academic advantages. PBL teaches students to think critically, communicate effectively, and apply classroom knowledge to unscripted, authentic problems. It allows undergraduates to frame, research, and investigate complex real-world problems that span social and technological dimensions for nonprofit organizations and sponsors (Elmes & Loiacono, 2009). CEL, on the other hand, directly connects and engages students with local communities. It involves participatory research that brings students and residents together to address shared issues, building social capital, deepening understanding of community needs, and strengthening collective adaptation in

challenging environments (Downs et al., 2009). Together, these experiential approaches demonstrate how hands-on learning links academic study with real-world impact, fostering both community engagement and student growth.

**Figure 1**

*Characteristics of a high-impact practice.*



*Note.* Adapted from *Implementation of high-impact practices in engineering design courses* by D. Pusca & D. Northwood, 2018, *World Transactions on Engineering and Technology Education*, 16, 108.

The value of these approaches is proven in their ability to address complex real-world problems while contributing to broader societal goals, specifically through the approach of CEL. For instance, Stephens, Hernandez, and Boyle (2009) describe how collaboration between universities, local government, and research councils in Argentina led to the creation of a Science and Technology Pole, demonstrating how academic–community partnerships can strengthen organizational capacity, innovation, and regional development. This stands as a prime illustration of why CEL consistently produces some of the strongest academic and personal growth outcomes for students (Pusca & Northwood, 2018). But what does this approach offer the communities themselves? CEL supports economic development, improves

infrastructure, enhances education and health initiatives, and strengthens neighborhood and school improvement efforts—from installing solar panels to staffing community health centers (Downs et al., 2009). Another clear example comes from CEL placements across urban and rural communities in three South African provinces, where students helped reduce clinic overcrowding and improve access to care (Diab & Flack, 2013). By applying classroom knowledge, students examined patients locally, preventing many from traveling long distances to hospitals or waiting overnight in long lines. Their presence eased hospital congestion and reflected genuine attention to community needs and beliefs. As patients in Limpopo expressed, “they checked me more than the sisters do” and “they did not rush anyone; they took a long time checking me” (Diab & Flack, 2013, p.3). Because of this approach, community leaders and residents viewed the students positively, recognizing that they truly listened. Through this experience, students developed key professional and interpersonal skills such as patient observation, clear and empathetic communication, active listening, and community-engaged problem solving. More broadly, CEL fosters sustainable projects that integrate academic learning with real-world application, allowing students to contribute their skills, knowledge, and technical expertise to community challenges (Stephens, Hernandez, & Boyle, 2009).

While these partnerships and high-impact learning methods provide substantial benefits, they also introduce challenges that can complicate collaboration and limit overall impact. Sustaining high-impact learning depends on time, trust, and genuine partnership. Reilly, Langley-Turnbaugh, and Neikirk (2023) argue that students often report feeling limited by time constraints, believing they cannot make a meaningful or lasting difference within short project timelines. Community organizations, meanwhile, prefer longer commitments (ideally a semester or more) so their investment of time is worthwhile and so students can fully understand the partner’s mission, operations, and needs (Downs et al., 2010). This deeper engagement allows student contributions to create a longer-term impact. However, even with these intentions, partnerships can face obstacles. If a community partner has a negative experience with a university or if the university has a poor local reputation, trust can weaken, making engagement more challenging (Reilly, Langley-Turnbaugh, & Neikirk, 2023). Building and maintaining trust requires ensuring that all parties have access to resources and share aligned goals and expectations. Yet these relationships can become hard to navigate when roles are unclear or when the outcomes of collaboration are uncertain. True collaboration requires clear goals, consistent communication, and mutual understanding between all partners (Downs et al., 2010). In essence, community-engaged learning can create meaningful benefits for both students and

communities; however, understanding whether these efforts yield the intended outcomes requires careful evaluation. Community impact does not disappear when a project ends, but without significant documentation, it can overlook the meaningful work and relationships built between universities and community organizations. To ensure that community partnerships remain effective, sustainable, and responsive to local needs, we now turn to the importance of measuring impact.

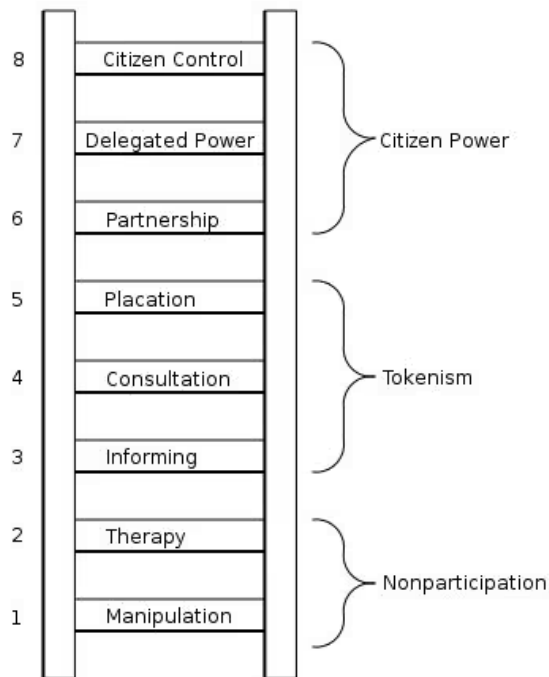
## **Measuring Impact**

Understanding a program's effectiveness is important to ensure it best serves a targeted community. Measuring impact refers to systematically assessing whether a project creates the changes it intends to, how those changes occur, and the extent to which the project itself is responsible for them (Gertler et al., 2016).

To effectively and properly assess the impact of a project, the project needs to have a descriptive agenda. A descriptive agenda describes the intended outcomes of a project and the methods and procedures used to accomplish the desired outcomes. Such a preemptive approach allows researchers and the community to make suggestions to improve a program for future implementation (Gertler et al., 2016). These iterative adjustments make a project more effective at accomplishing desired outcomes and can also reshape project goals to better suit the community's needs (Michel & Schneider, 2025; Gertler et al., 2016). Simply stating the goals of a project and analyzing how well the project accomplished those goals is not enough to help a community understand the effectiveness of a project (Gertler et al., 2016). This process of community involvement in shaping goals and outcomes can be understood through Arnstein's (1969) Ladder of Citizen Participation (see Figure 2). The ladder illustrates that true effectiveness and legitimacy come from moving beyond merely informing or consulting communities toward genuine Partnership and Citizen Control, where the community holds real decision-making power.

**Figure 2**

*Arnstein's Ladder of Citizen Participation.*



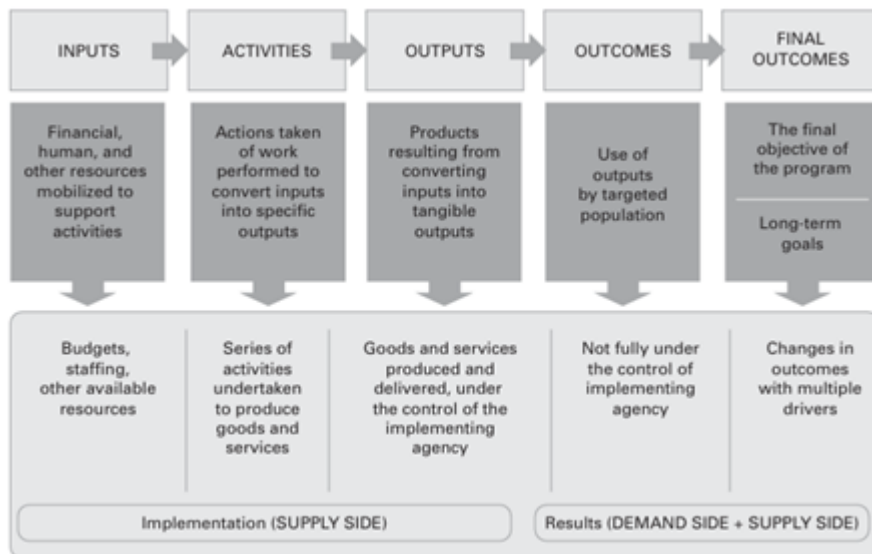
*Note.* Visual representation adapted from Arnstein (1969).

The project also requires a preliminary analysis of how the program will theoretically accomplish desired results (Gertler et al., 2016). A Theory of Change (ToC) is a description of how a program is supposed to accomplish desired results (see Figure 3 below) (Michel & Schneider, 2025; Gertler et al., 2016). A good ToC can provide room for a detailed analysis of not only the final results but also all of the resulting details used to accomplish desired outcomes.



**Figure 3**

*Results Chain that implements Theory of Change (ToC)*



*Note:* This is a diagram that describes the resources, activities, and allocation of resources that serve as a plan to accomplish a desired outcome.

One way to analyze the impact of a program is the Community Capital Framework (CCF). The CCF provides 7 different categories of community capital, or categories that describe how healthy a community is (Figure 4) (Chazdon et al., 2017, Greene et al., 2022). The CCF is often used in custom procedures for assessing the big picture of the impact of a program or intervention in question.

How could someone assess the impact of a community garden on neighborhood pride? One way to assess impact is using Ripple Effects Mapping (REM). REM is an impact evaluation method that examines how assets could affect other assets (Figure 5). An asset is a piece of qualitative data that can affect one or more categories of community capital (Chazdon et al. 2017; Greene et al., 2022). In the context of a community garden, one example of an asset is that the establishment of a community garden causes people to socialize more. Another example of an asset is that socializing causes the improvement of well-being. REM examines direct effects of a project causing other indirect effects (Chazdon et al., 2017, Harris, 2024). Socializing is a direct effect of the establishment of a community garden, while the improvement of well-being is an indirect effect. REM presents a visual and pragmatic way to analyze impact, making it easier for stakeholders and community members to engage in impact analysis alongside researchers (Chazdon et al., 2017; Greene et al., 2022; Harris, 2024).

**Figure 4**

*Community Capital Framework (CCF)*

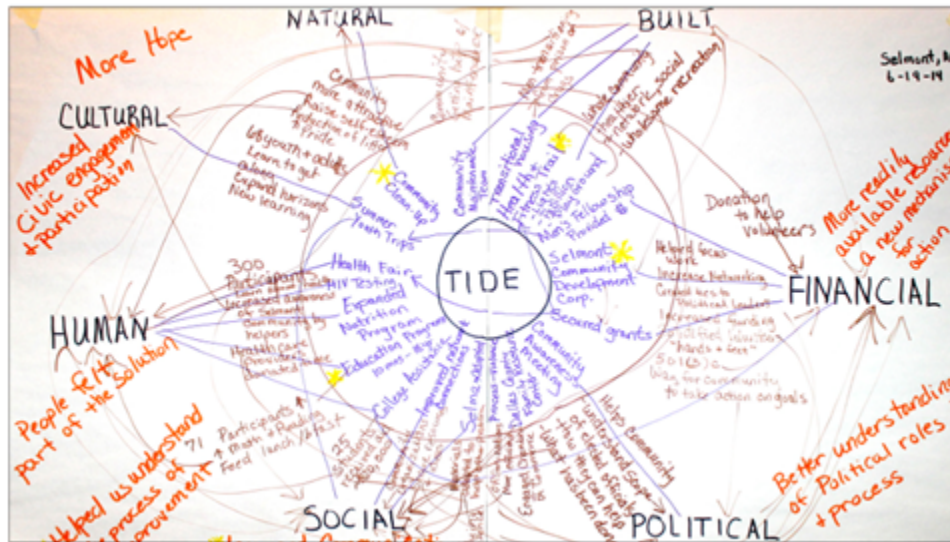


*Note:* A community capital describes how healthy a specific aspect of a community is.

The community garden to socializing and improving well-being, resembles a visual chain of effects. One limitation of REM, however, is that REM only provides a surface-level analysis of the relationships between effects that are both directly or indirectly caused by a project. It is helpful to ask why an effect causes another effect. In order to understand the effects of a community garden, one can reveal more by asking why the community garden causes people to socialize more and why socializing improves well-being. Another method for impact evaluation that examines a causal relationship is Realist Evaluation (Harris, 2024). Realist Evaluation is used to examine the hidden effects caused by a program. REM and Realist Evaluation can be used together as a method called Realist Ripple Effects Mapping (RREM) to examine beyond the surface-level effects observed by REM. In essence, these measurement tools not only reveal what changes occur within a community but also provide the evidence needed to communicate those changes clearly and credibly.

**Figure 5**

*Ripple Effects Map of Turning the Tide on Poverty (Tide)*



*Note:* Tide on Poverty (Tide) is a project that aims to help local citizens make their voices heard in their local communities, with a topic of dialogue being about poverty.

People need more than quantitative data showing improvements in community health due to a project; they also require evaluations that explain the extent to which the program contributes to the changes observed (Michel & Schneider, 2025; Gertler et al., 2016). Another goal of impact evaluation is to help improve the quality and efficiency of a program (Gertler et al., 2016). Because these projects are increasingly complex, people have a harder time staying up to date on their progress, which leads them to question how effective they truly are (Michel & Schneider, 2025). For that reason, the demand for impact evaluation is high and will continue to increase for the foreseeable future (Michel & Schneider, 2025; Chazdon et al., 2017). Many programs, particularly non-profit organizations, require external funding to continue running, which requires community support. So one goal of impact evaluation is to maintain public support for a program (Michel & Schneider, 2025).

With the CCF to map assets to community capitals and the visuals to display how assets affect other assets from REM, intended and unintended changes can be revealed from impact analysis (Chazdon et al., 2017). By understanding how and why impact happens, projects are better positioned to share their results in ways that build trust and engagement. This connection between evaluation and communication sets the foundation for promoting a project's work effectively, which the next section explores.

## Promoting and communicating outcomes

Once a program is assessed, the next step is to communicate the results to the broader community. Strong measurement tools help evaluate and present outcomes, but they remain secondary to the core priority: fostering authentic, meaningful participation. To bridge assessment and communication, a project needs to systematize an evidence-based record, moving beyond vague assertions to maintain a detailed "impact log" of specific, tangible outcomes (Hough, 1984). This demands a strategic communication framework that goes beyond simple campaigning toward authentic public engagement. This framework includes four parts for effective public outreach: first, setting a strategy based on audience insights; second, translating data into a compelling narrative; third, validating that narrative with evidence; and finally, leveraging this foundation to build a durable community legacy (Benford & Snow, 2000; Reilly, Langley-Turnbaugh, & Neikirk, 2023). This disciplined documentation serves a dual purpose: to provide the raw data for formal impact measurement and, more importantly, form the undeniable foundation for persuasive public claims about a project's success, thereby securing community trust and legitimizing the entire endeavor (Downs et al., 2009; Gertler & World Bank, 2016; Michel & Schneider, 2025).

Understanding community participants forms the foundational task. It is important to gain a deep, empathetic understanding of the beliefs, drivers, and lived experiences of community participants, which are the primary data for any effective initiative. Andreassen's (2005) "marketing mindset" is crucial here, as it treats audience understanding not as an afterthought, but as the core of design. This discipline helps avoid strategies built on internal assumptions, which inevitably lead to participant mismatch and poor outcomes. Instead of internal assumptions, it's important to apply this "marketing mindset" to other key audiences as well: the university (seeking validation) and potential funders (seeking accountability and scalable results). For these audiences, measurement tools such as surveys and interviews translate community insights into formats that outsiders can easily understand (Andreassen, 2005; Schensul & LeCompte, 2013). These tools help convert deep, qualitative work into the numbers and clear explanations sponsors expect when reviewing impact (Downs et al., 2009; Gertler & World Bank, 2016; Michel & Schneider, 2025). Therefore, participatory work follows two paths: engaging the community while collecting supporting measurements. Measurement tools should be used to record and learn from the process, not to control it. This approach ensures the work stays meaningful for the community while providing information that supports funding and institutional backing.

After understanding the community, the next major step is to frame the project's message. Storytelling plays a central role here. Connecting the project's message to the community's core beliefs and emotions (Benford & Snow, 2000) makes raw data interviews, surveys, statistics, persuasive, and personally relevant. The narrative captures public attention and motivates participation; therefore, creating an impactful storyline and using behavioral tools to measure engagement is essential. A story's legitimacy and verifiable evidence build the trust needed to transform a compelling narrative into a credible claim.

The ultimate ambition of this strategic communication process is to leverage resonant storytelling and documented credibility to evolve the project beyond a single initiative into a durable community institution (Andreasen, 2005; Downs et al., 2009; Reilly, Langley-Turnbaugh, & Neikirk, 2023). In terms of Arnstein's Ladder, effective communication is the essential mechanism for ascending from the lower rungs of Informing and Consultation toward the higher levels of Partnership and Citizen Control. By synthesizing consistent and meaningful public engagement, the project cultivates a strong, trusted "brand" that acts as a promise for quality and reliability, fostering long-term support and allegiance (Eshuis, 2012). Making trust and collaboration visible is essential for proving that the project is becoming a real community institution. Creating visuals like stakeholder maps to show how different groups are connected, and timeline charts highlight when community feedback turns into action. Tools like social network diagrams measure the strength of these partnerships. These visuals turn hard-to-see relationships into clear evidence, showing that the project is moving up Arnstein's Ladder toward true partnership and strengthening its trusted reputation in the process (Ware, 2013).

This transformation is critical, as it shifts the project's identity from a temporary intervention to a permanent fixture, thereby ensuring its long-lasting influence and embedding it as a self-sustaining force within the community. In essence, the path to profound and measurable public impact is not a matter of chance but of a deliberate, integrated strategy. By first grounding the approach in a deep understanding of the audience, then translating technical outcomes into a resonant narrative, rigorously supporting that narrative with verifiable evidence, and ultimately leveraging this credibility to build a trusted brand, a project transcends its immediate goals (Reilly, Langley-Turnbaugh, & Neikirk, 2023). This cohesive process ensures that initial engagement matures into lasting behavioral change and cements the initiative's legacy as a durable and indispensable community institution.

This change is crucial because it turns the organization's identity from something temporary into a permanent fixture. This ensures its influence continues and roots it within the community as a self-sustaining force. In essence, achieving a deep and measurable public impact does not happen by accident. It requires a careful, connected plan. The process starts by building a strategy on a solid understanding of the audience. It then turns project results into a story people connect with, backs up that story with hard evidence, and finally uses that credibility to create a trusted brand. To fully support this shift, the communication strategy also requires deliberate promotion tactics that help the story spread and build momentum. Alongside strong messaging, the project must activate channels that maximize visibility—such as leveraging social media platforms, partnering with local organizations, and encouraging word-of-mouth sharing through community leaders and participants. This is how a project moves beyond its initial goals. This unified process makes sure that early involvement grows into lasting change and secures the project's legacy as a strong and essential community institution.

## **Our Project**

This project is in collaboration with Worcester Community Project Center (WCPC) and our sponsor, Amanda Wittman, who is an assistant teaching professor in the Department of Global and Integrative Studies at Worcester Polytechnic Institute. Our team and Amanda focused on developing data-driven storytelling and exploring innovative ways to describe what makes meaningful change. By combining impact assessment with narrative techniques, we capture both the outcomes of community-engaged learning and stories that illustrate its importance, providing a promotive storytelling framework that captures and communicates the 25-year legacy of the WCPC's community-engaged learning.

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# Methods

The goal of our project was to assist the Worcester Community Project Center (WCPC) in documenting the stories and impact on individuals and communities. Below guided us in selecting effective strategies and methods to measure the WCPC's impact on the greater Worcester area.

Our objectives were to:

- **Identify** the different themes and topics of WCPC.
- **Understand** the impact of the WCPC on a key aspect of the greater Worcester Community
- **Develop** and get feedback on promotional materials

This helped us ensure that we established clear plans for promoting and sharing the data we collect, and that any outreach efforts were supported and validated by the community.

## Objective 1: Identify key themes and topics

Our first objective was to develop a deep understanding of Worcester Community Project Center (WCPC) projects—their goals, accomplishments, challenges, and overall impact on the community through their community-engaged learning experiences. To achieve this, we identified the major themes and topics represented in WCPC projects over the past twenty-five years. This understanding guided our approach to analyzing patterns and relationships that reveal how WCPC projects have shaped the Worcester community.

### *Content Analysis*

To accomplish this objective, we conducted a content analysis of the WCPC website and the Gordon Library database. We reviewed past projects, including their themes, partner organizations, and focus areas, to identify recurring topics that appeared throughout the project archive (Columbia University Mailman School of Public Health, 2026). Initially, we attempted to use convenience sampling by selecting a small subset of projects from the WCPC website to begin our analysis. We focused on projects categorized under the UN Sustainable Development Goals of Zero Hunger and Quality Education and analyzed several from each category. However, we realized that this approach did not provide a comprehensive understanding of WCPC's overall impact. Analyzing only 20 projects out of 268 did not allow us to fully grasp the

breadth of the Center's goals, themes, and accomplishments. As a result, we revised our approach and decided to analyze all 268 projects. To ensure consistency and efficiency, we first evaluated one project together to establish a shared understanding of how each project should be analyzed and coded, which is a process known as norming (Caprar, 2012). Norming helped us standardize our process and maintain consistency across our data collection. We reviewed each project's abstract, acknowledgements, and title pages and recorded relevant information in an Excel spreadsheet. We organized the data using the following categories:

- **Project Name** – the official title of the project
- **Sponsor** – the individual who supported and guided the project
- **Organization** – the partner organization involved
- **Organization Keywords** – a brief description of the organization's role in the project
- **Project Type** – the primary function or purpose of the project
- **Project Goal** – the overall objective of the project
- **Year** – the year the project was completed
- **Theme** – the corresponding UN Sustainable Development Goal associated with the project

After we finalized our coding structure, we divided the 268 projects evenly among the three team members and assigned 89 projects to each person. We color-coded the spreadsheet to clearly indicate project assignments and ensure that no project was analyzed more than once. Once we completed the content analysis of all 268 projects, we prepared it for further analysis by examining the relationships among the coded categories to identify patterns, recurring themes, and connections across all projects.

Using the completed spreadsheet as our master database, we applied a funnel technique to produce a series of additional tables to examine relationships between categories, letting us fixate on finer links and connections. For example, we created a pivot table derived from our content analysis data that calculated how many times each organization appeared across all projects. This required counting recurring organization names within the dataset to show repeated partnerships. We then focused on the Organization Keywords column. Because many keywords were highly specific, we grouped related terms into broader thematic categories. To do this, we reviewed all keywords, identified patterns, and clustered similar terms under shared umbrella themes. We color-coded the spreadsheet to visually group keywords into these broader categories and added short descriptions for each theme to maintain clarity. After



grouping the keywords, we counted how many organizations fell under each broader theme. Then we organized and categorized the content analysis data, and we used the structured dataset to support network mapping. The coded spreadsheet served as the foundation for creating visual representations of relationships between projects, organizations, and themes.

In essence, combining content analysis with network mapping strengthened our ability to understand WCPC's impact. The content analysis identified the key themes, organizations, and project characteristics (see Appendix A), while mapping translated these findings into a visual representation that reveals how projects fit together across time, organizations, and community networks, as well as the results we got from our content analysis and other pivot tables. This method approach helped us see not just what the WCPC has done, but how its work has shaped relationships, strengthened partnerships, and supported different neighborhoods over the past twenty-five years. By visually layering themes, locations, organizations, and project relationships, we developed a clearer and more accessible understanding of the WCPC's long-term influence, fulfilling the core purpose of Objective 1.

## **Objective 2: Understanding the Impact**

After gaining an understanding of the themes and partnerships of the programs and interventions of the WCPC, we needed to better understand how such projects and interventions have impacted the community. Thus, our second objective was to perform an impact assessment on the programs the WCPC has been involved in. We also needed to gain an understanding of the connections between organizations in partnership with the WCPC. By completing our second objective, we obtained qualitative data that thoroughly describes the impact of a given program on one or more themes. To obtain the qualitative data that rigorously highlights the impact of a given program, we conducted semi-structured interviews with sponsors. We then used Ripple Effects Mapping (REM) to describe impacted members of the Worcester community in the form of chains of effects that resemble a ripple.

### *Inviting Community Leaders to Interview*

Once we selected the derived themes for our content analysis, we began conducting one-on-one semi-structured interviews with sponsors. Before scheduling interviews, we systematically gathered contact information for each project sponsor by searching organization websites to locate contact information. Once we obtained available contact information for sponsors within each theme, we emailed all reachable sponsors with an invitation to participate.

For each invitation, we explained the purpose of the interview and clarified that the conversation would focus on a project they previously worked on to assess its impact on a specific theme (see Appendix B). To avoid sending duplicate invitations, we coordinated using a shared Google document listing all candidate emails. For each sponsor, we set and booked coordinated interview times and recorded scheduled meetings on Calendar Booking, shared with the whole team.

### *Interviews*

To collect the qualitative data necessary for Ripple Effects Mapping (REM), we conducted 11 semi-structured interviews with 10 organizations (see Appendix C). These participants provided broad overviews of project impact, connections between organizations, strategic influence, and personal experiences. Semi-structured interviews allowed us to guide the discussion toward our research objectives while remaining open to participants' unique insights, ensuring the collection of focused and relevant qualitative data (Greene, Stroepe, & Holston, 2022). Each interview lasted approximately 20–30 minutes (see Appendix D). The semi-structured format balanced guided questions with flexibility, enabling participants to share detailed stories and reflections. With participants' consent, we audio-recorded each interview, took written notes during the conversation, and later transcribed the recordings to support deeper analysis.

These interviews served as the primary method for collecting rich, narrative data. They provided essential qualitative evidence to assess program impact and to understand the complex challenges faced by communities. After transcription, we conducted an initial thematic analysis to identify key effects, relationships, and patterns. This analysis formed the foundation for our REM process and supported an evidence-based impact assessment.

### *Ripple Effects Mapping*

During each interview, we asked questions (see Appendix D) that fulfilled the appreciative inquiry process of REM (Chazdon et al., 2017). With these questions, we learned more than just about the immediate effects. We also learned about many of the long-term and indirect effects. An effect refers to an outcome that results from the project being assessed and how it is connected to multiple other outcomes and perspectives (Harris, 2024). From each interview, we also examined the causal relationships between the effects identified, but for statements, and any significant ripple impacts that were observed. We strengthened our

understanding of these connections through sponsor insights, which allowed us to develop a more nuanced explanation of how these effects interacted (Chazdon et al., 2017). From the interview transcriptions, we summarized what we learned from the interviews, including direct effects, indirect effects, interactions with other organizations, and some of the challenges faced when making the program in question happen. With a clear overview of the direct and indirect/long-term effects for each interview, we examined the ripples of effects that we learned from all the interviews, noting any details about program impact in common across organizations. We analyzed the ripples of effects we learned by adding descriptions of impact from several different project examples along with the visualizations we made in objective one. Visualizations in particular included the derived organization themes and the project themes that described what student researchers did specifically for each WCPC project. Once we finished analyzing what we learned from each interview, we were prepared to package and communicate the insights we learned from program sponsors to a broader audience.

### **Objective 3: Develop and get feedback on promotional materials**

We conducted 2 feedback sessions, one with the Director of the Worcester Community Project Center, and another with our sponsor, Amanda Wittman, with our advisor Courtney Kurlanska to ensure that our content aligned with their understanding of the WCPC's mission (see Appendix E). Our goal was to make the materials as storytelling and personified as possible while accurately reflecting the human stories behind the work.

Each feedback session lasted approximately 30 minutes. During these sessions, we presented our deliverable website to see if our visualizations clearly and accurately represented the WCPC's impact. As well as its usability, accessibility, and overall effectiveness in communicating the information we had found. These discussions revealed not only what participants thought but also the reasoning and emotions behind their interpretations. They provided feedback on what to add or revise and whether our data visualizations clearly and effectively communicated their intended purpose. During each feedback session, we wrote notes about any areas of improvement for our deliverables. We used this feedback to revise our website, making it clearer and easier to understand, while also ensuring that all information presented was accurate and well supported.

### *Object Response Testing*

Object response testing complemented the feedback sessions because it captured participants' immediate reactions to the individual materials. Using this method, we understood first impressions, what captured attention, what was confusing, and what elements resonated on a quick, instinctive level. We presented each page of the website one at a time and asked for their first impressions on what captured attention, what was confusing, and what resonated (Appendix E). These fast and rapid responses helped us identify whether the materials were communicated effectively at a glance, which was essential for public-facing communication. Thus, we wanted to ensure that the first impression reactions and products we presented were easily understood and relatable to our focus groups. Following both methods helped us confirm key adjustments before finalizing the promotional materials.

In essence, combining feedback sessions with object response testing strengthened our ability to refine promotional materials that accurately and compellingly represent the WCPC's work. Each method provided a different layer of insight with deeper interpretation through group discussion and immediate interpretation through rapid response. Together, they have ensured the final materials are clear, engaging, and grounded in the community's perspectives, fulfilling the purpose of our last objective.

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# Findings

The following chapter covers and presents the key findings from research, coded data, and conducted interviews with partners and sponsors. These findings helped symbolize the patterns and impact of the Worcester Community Project Center (WCPC), and ensured that our promotional materials are accessible and show the effect that WCPC has on a key aspect of the Greater Worcester Area.

## Finding 1:

**Over the last 25 years, the Worcester Community Project Center has helped 168 organizations with student projects (Appendix G).** Based on our content analysis, the most common areas these organizations worked in were grouped into themes that emerged as the most recurrent across all organization keywords (Map 1):

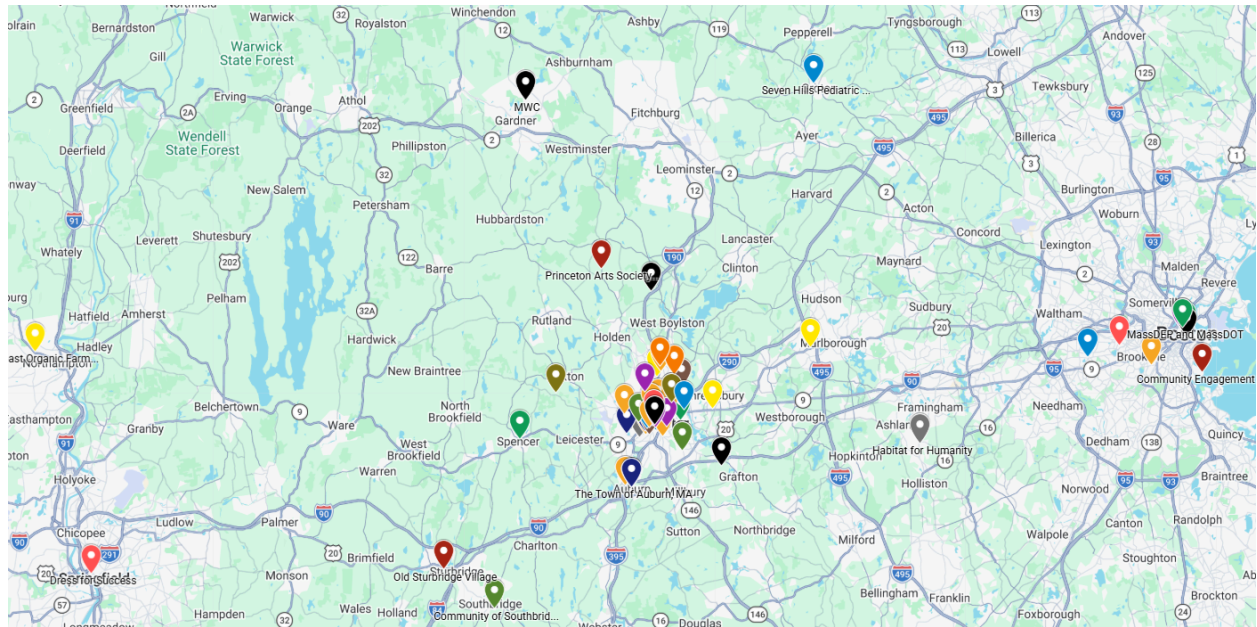
- Environment & Sustainability
- Education & Learning
- Community Development.

Fifty organizations focused on Environmental & Sustainability issues, including areas such as climate and renewable energy, conservation, water, and waste. Thirty-eight organizations focused on Education & Learning issues, with efforts centered on academic institutions, STEAM, literacy, and curriculum development. Lastly, thirty-four organizations focused on Community Development, addressing areas like civic engagement, volunteering, and neighborhood involvement (Appendix F). These numbers highlight the primary areas in which WCPC has contributed sustained, long-term value within the Worcester community.

**Geographical mapping also highlights WCPC's strong community presence.** Most partner organizations are located near downtown Worcester or around WPI's campus, demonstrating WCPC's deep local connections and also easy accessibility for students and community members. While the majority of partners are within Worcester County, some of these organizations' headquarters or offices are located in the Boston area (Map 1). This shows how some connections can travel as far beyond the immediate and close neighborhood boundaries of the Greater Worcester Area, and highlights WCPC's targeted reach beyond Worcester.

## Map 1

### *Geographic Organization Map*

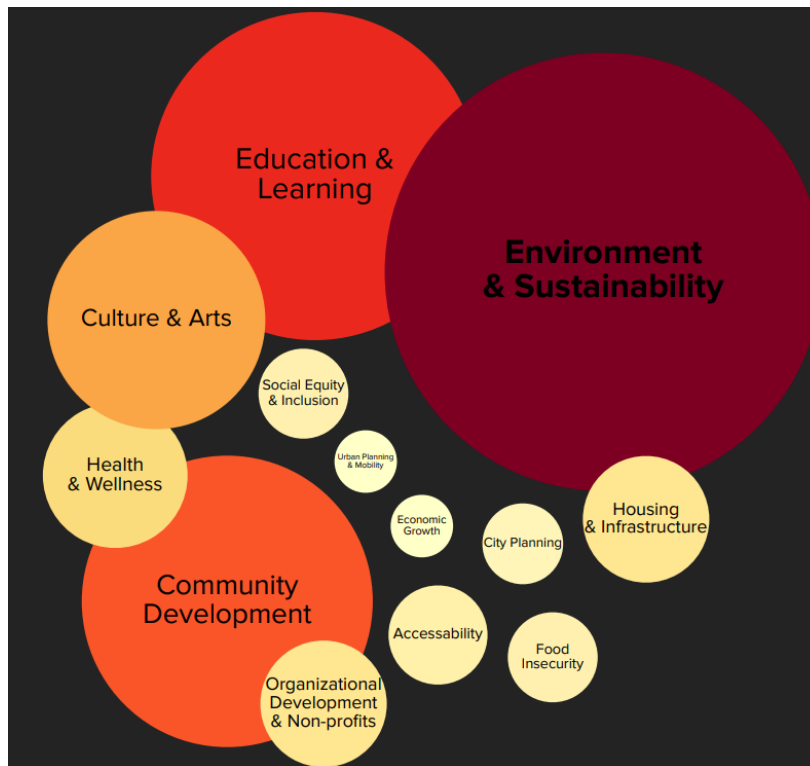


*Note.* A geographic map of organizations that worked with WCPC was created in Google Maps.

Link to map [HERE](#)

## Map 2

### Heat Map of Organization Themes



*Note. Heat map made from the software Kumu. Click [HERE](#) for the link to the map*

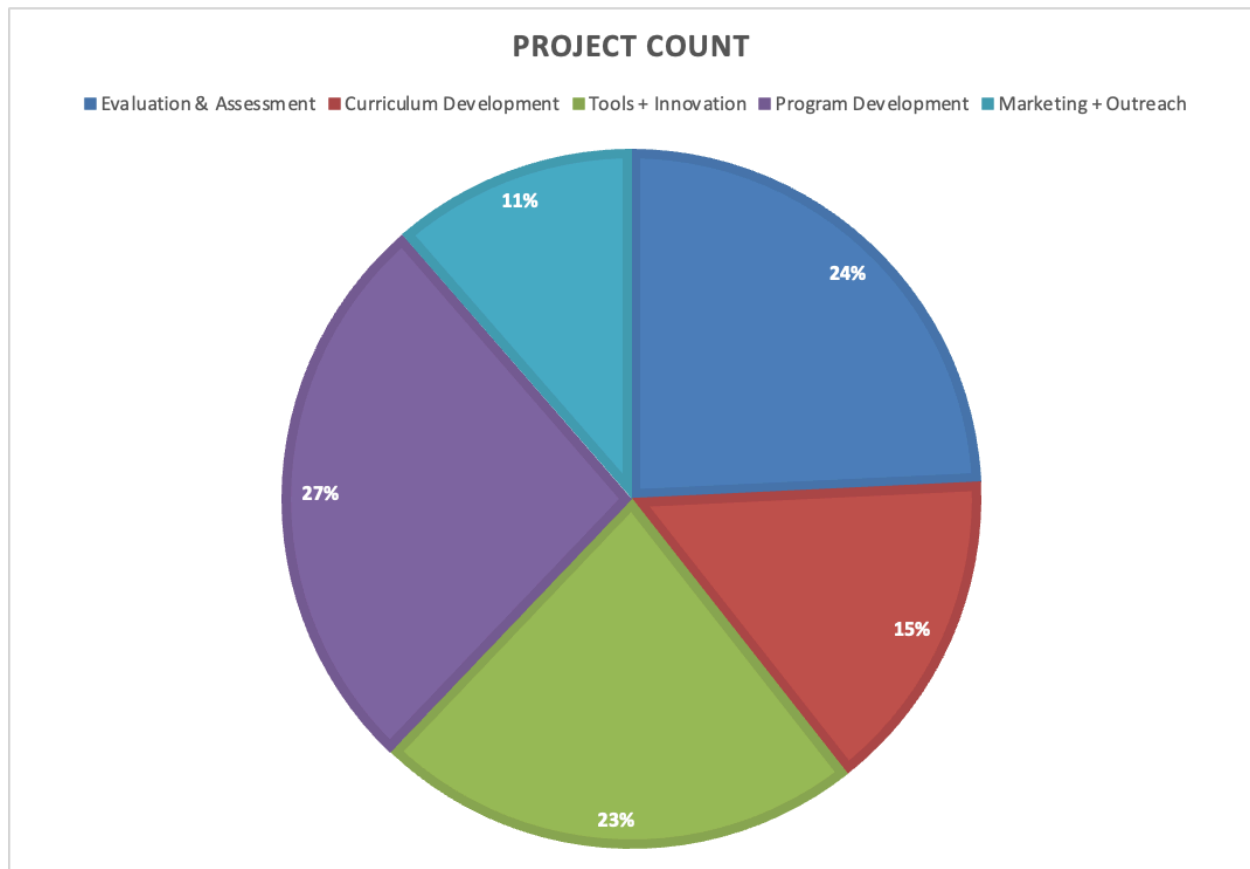
This theme map helped visualize how often organizations engaged with each theme and illustrates the WCPC's strong, repeated involvement in environmental issues, educational access, and partnerships with these organizations, which reflect local needs like environmental challenges in Worcester, ongoing educational access work, and neighborhood development. "In 2019, the city of Worcester declared a climate emergency, stating that climate change threatens the community's environment" (Shrestha, 2021). These consistent themes demonstrate WCPC's role in supporting the organizations and communities working to strengthen Worcester.

However, we also wanted to understand *how* organizations actually used WCPC projects to advance the challenges and issues they were addressing. We found that **most projects, 27%, focused on program development** (Figure 6), with student teams partnering with sponsors to create or improve initiatives such as food systems,<sup>1</sup> design implementations,<sup>2</sup> and climate-planning tools that help organizations deliver services more effectively.<sup>3</sup> This reflects the core structure of community-based learning, where students apply classroom knowledge to

build practical systems, tools, and resources that strengthen organizational capacity (Elmes & Loiacono, 2009).

**Figure 6**

Project Themes Per Count



*Note.* Shows the distribution of project themes by count deprived from our Content Analysis Excel sheet.

Our interviews provide concrete examples of how this capacity building occurred. For example, a project with Habitat for Humanity developed a standardized tonnage calculation model that is now used across ReStores nationally, strengthening environmental reporting and improving access to grants.<sup>4</sup> A translation tool created for Habitat ReStores enhanced communication with Spanish-speaking staff and donors, expanding accessibility and community engagement. An assistive technology database developed for Seven Hills streamlined how staff identify and recommend apps, improving service delivery for individuals with disabilities.



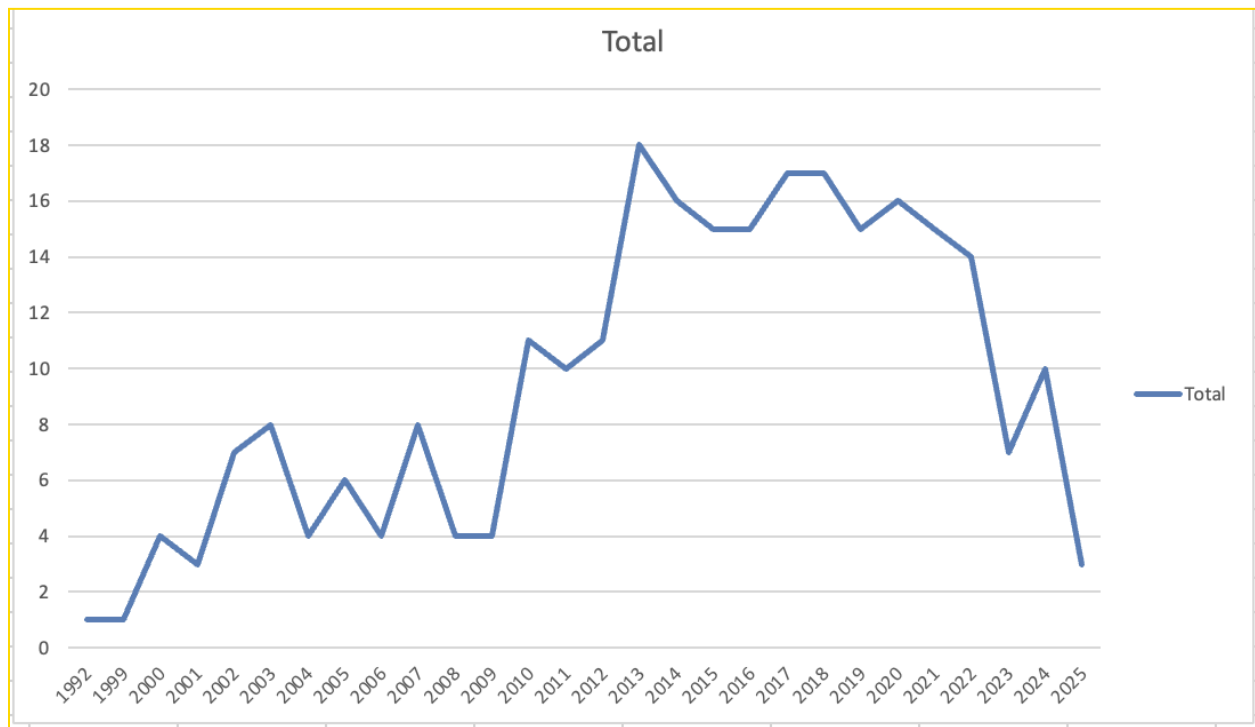
Similarly, a media storytelling project helped a nonprofit better communicate its mission and impact, strengthening community awareness and outreach.<sup>5</sup>

Together, these examples demonstrate that WCPC projects do more than complete short-term tasks; they create lasting infrastructure that organizations continue to use, reinforcing the long-term impact of community-based learning within Worcester. This kind of work benefits a variety of organizations, which is why we see similar capacity-building projects appear across organizations working in different thematic areas. For example, Habitat for Humanity addresses eight different themes, representing 61.5% of all identified themes. Many of these efforts support long-term sustainability initiatives within the Worcester community.

To understand **how the WCPC's involvement in the community evolved over time**, we also examined the number of projects completed each year. While the thematic distribution shows what areas WCPC has supported most, the annual project count helps reveal when this work occurred and whether WCPC's activity has changed across its 25-year history. This view gave us a better understanding of the Center's overall stability, periods of growth, and recent declines that were influenced by external factors.

**Figure 7**

*Project Count Per Year Graph*



*Note.* The vertical axis refers to project count, while the horizontal axis refers to project year.

We created a graph that visualizes the number of projects the WCPC has been involved in each year (see Figure 7 above). The distribution of projects is clearly non-uniform. Since the establishment of the WCPC in 2000, annual project activity generally either increased or remained steady from year to year. However, beginning in 2020, the number of WCPC projects has declined sharply.

In an interview with the director of projects at the WCPC, we found that this decline is largely connected to the introduction of the \$5000 Global IQP Scholarship, which made international IQPs much more accessible for students. Because WCPC projects cannot compete with the appeal of a fully funded international experience, student interest in completing their off-campus IQP in Worcester decreased. The scholarship went into effect in 2018, but the impact on WCPC project numbers did not appear immediately. This delay is explained by the COVID-19 lockdown in 2020, which temporarily halted international travel and paused the shift toward global IQPs. Once travel reopened, the effect of the scholarship became visible, and WCPC project numbers began to decline at a faster rate. Another contributing factor

was that at the beginning, students were given the difference in cost, but this amount was later reduced.

Taken together, this illustration of different locations shows that the WCPC serves as a guiding anchor for local and Worcester-based organizations. Its long-standing credibility allows it to attract partners across the region, while its sustained collaborations demonstrate that WCPC's influence extends well beyond single projects, supporting a networked, multi-year impact across the Greater Worcester Area.

## Finding 2:

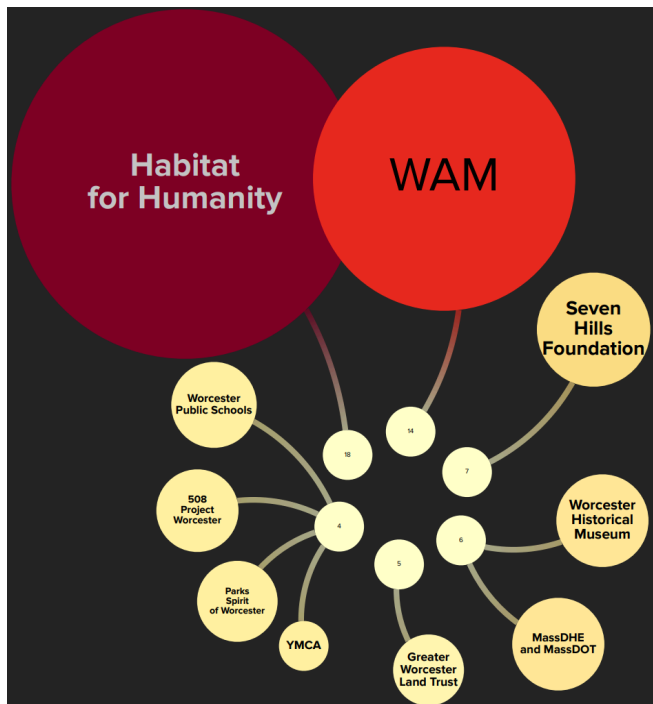
**Worcester Community Project Center (WCPC) creates long-term relationships with real-world impact.** These collaborations reflect not only the Center's reach across different sectors but also its ability to maintain trust-based relationships over time. Within the 168 organizations that the WCPC has supported across a diverse range of thematic areas. Within these different organizations 50 partnered with the center on environmental initiatives, 38 on education, and 34 on community development.

Three organizations stand out for their sustained engagement with WCPC: Habitat for Humanity, Worcester Art Museum, and Seven Hills Foundation. As shown in Map 2, these partners account for 12% of all documented collaborations since the Center's establishment in 2000. These recurrent contacts show a deeply rooted institutional network rather than one-time engagements, and these frequent partners usually connect with other organizations across project themes; indirectly, WCPC projects create networks of non-profits and collaborations.

The calculation model developed by students through an IQP project with Habitat for Humanity ReStore was used as a tool to quantify the environmental impact they had, and to create a stronger narrative to unlock grants and funding. That tool was then adopted by other organizations, ReStores across the region, nation, and country embraced it for its utility and simplicity and because of its major impact. Thus, it became a global tool that most ReStores rely on globally.

### Map 3

#### *Top Ten organizations by count*



*Note.* Visual representation of the organizations that had the highest count / Higher frequency.

Click [HERE](#) for the link to the map.

Through interviews with sponsors conducted as part of our second objective, we gained deeper insight into how these long-term partnerships function in practice. One common idea that emerged across several organizations was the importance of making data more accessible and improving communication, both internally and with the broader community. Many organizations described challenges related to inconsistent information, limited data organization, or difficulty communicating their impact to funders and the public. As a result, improving communication systems and strengthening data accessibility became a recurring need. WCPC projects often addressed this gap by creating tools,<sup>6</sup> models,<sup>7</sup> databases,<sup>8</sup> and storytelling resources that helped organizations better organize information and communicate their work more effectively.<sup>9</sup>

In addition, the Broad Meadow Brook Trail Development project demonstrates how partnerships can extend beyond a single organization. Through collaboration with Mass Audubon, students worked to create a physical and educational connection between the Broad

Meadow Brook Wildlife Sanctuary and the future Blackstone Valley Visitor Center. This project did more than simply map and blaze a trail; it connected conservation efforts, public education, and community access to nature.

Although we learned that the trail has not yet been formally completed, largely because projects of this scale require long-term coordination, land agreements, and continued development, the groundwork laid by the IQP team remains significant. The proposed route is already informally used by community members, demonstrating both its feasibility and public interest. The project therefore represents not only an immediate contribution, but also a foundation for future implementation that may be advanced in the coming years.

These efforts illustrate how WCPC projects do more than connect WPI students with a single sponsor. Instead, they foster sustained collaboration, strengthen institutional relationships, and create long-term community impact. Even when outcomes unfold gradually, the research, planning, and partnerships established through these projects contribute to a broader environmental and community network that continues to grow over time.

Efforts and work like the one done with Seven Hills Foundation symbolises how these projects do more than just connect WPI students with organizations and sponsors. Instead, they retain the sustained collaboration, and strengthen institutional relationships by creating long-term community impact. For instance, history projects with Seven Hills Foundation shows that over 67% of the WCPC engagement have been built directly from previous work. For example, the study on assistive technology Expo done in 2017 evolved into a Planning Guide and Toolkit a year later, giving the possibility to Seven Hills staff to host the event independently. This process demonstrates a type of partnership focused on long-term growth and continuous evolution.

### Finding 3:

**Organizations emphasized that WCPC projects help them address capacity gaps, such as staffing shortages and insufficient funding, by organizing projects and providing student research, which helps accomplish goals despite constraints.** Across five interviews, organizations mentioned limited bandwidth and a lack of sufficient funding as ongoing challenges.

For example, the Indian Lake Watershed Association (ILWA) described how one project focused on preserving decades of archival material, reinforcing the organization's long-term advocacy for water quality and invasive species management. Because ILWA operates with limited staffing, dedicating time to organize historical records was not feasible while also maintaining active environmental advocacy. By systematically organizing these records, the students helped bridge generational transitions in leadership and ensured that important contracts, strategies, and environmental data were not lost. Improved access to these archives strengthened the organization's institutional memory and continuity, directly expanding its capacity to sustain its work over time. These examples reflect a broader reality that many nonprofits operate within shifting funding and grant landscapes that change due to political and economic conditions, making sustained staffing and long-term planning even more difficult.

Through another interview with a non-profit organization named Abby's House, we found that they needed a way to digitize its artifacts for its 50th anniversary. However, the organization's primary mission is to provide shelter and services to women in need, and staff members are focused on that direct service delivery. While preserving their history and strengthening storytelling was important, they did not have the staffing capacity or time to dedicate to archival organization. Having previously worked with IQP students at Habitat for Humanity, the new volunteer coordinator sought to engage IQP teams. The students were able to dedicate sustained time and technical skills to digitizing and organizing the archives, allowing Abby's House to celebrate its anniversary and preserve its institutional memory without diverting staff away from their core services. In this way, the IQP filled a staffing and capacity gap that the organization could not address on its own.

Hence, the products that came out of these projects are innovative and solve real-life issues. Organizations and the director of projects at the WCPC have also described some of the challenges organizations experience. *"All of our project sponsors were just dealing with really difficult work and life situations themselves. A lot of our organizations serve the community, and they were serving people who had lost their jobs. Food banks were suddenly flooded with people who needed assistance. Some organizations closed, some were just dealing with a lot. It was just a very difficult time for the community."* Such issues were especially exacerbated by the negative effects of the COVID-19 lockdown. Student researchers helped the WPI STEM Education Center systematically track community engagement activities. Part of this project included learning how to use this tool called Collaboratory. Collaboratory is a software that is used to help track community engagement. While this software is quite powerful and verbose,

the verbosity makes Collaboratory a challenging tool to learn how to use. Staff at the WPI STEM Education Center did not have the time to learn how to use Collaboratory. Student researchers in the project helped the WPI STEM Education Center to finally get started with Collaboratory which ultimately improved their ability to track engagement from then on.

We also learned from organizations that collaborating with the WCPC benefits not only student learning and professional experience, but also the sponsoring organizations. Student researchers have developed practical solutions that strengthened collaboration between organizations and the WCPC. For example, student researchers worked with the organization HECCMA on a project to connect college campuses to the Worcester community. With this goal, they created a centralized event calendar that allowed several universities to post upcoming public events in one shared space. The founding director of HECCMA described the process the event calendar does to retrieve public university events from university websites as “scraping”. This “scraping” mechanism is a solution one of the student researchers came up with to automate the process of adding university public events. This shows that student researchers help organizations come up with new ways to solve problems. This calendar, although short-lived, helped improve engagement and community outreach between the wider community and universities.

Similarly, Habitat for Humanity worked with student researchers to develop a translation tool briefly mentioned in finding 1. This tool’s purpose was to improve communication between Spanish-speaking staff, donors, and customers. It would help make it easier and more inclusive for interaction between the ReStore and community members and donors, providing little to no room for misunderstandings caused by language barriers. While improving communication was one outcome, the deeper impact was on organizational capacity. Staff members were previously spending additional time navigating language barriers, which slowed transactions and limited engagement with Spanish-speaking donors and customers. The IQP students were able to apply technical skills and dedicate focused project time to designing a structured solution. As one sponsor explained, *“It changed the relationship between the shoppers, the donors, and our organization. People came in, and they could communicate. When they left, they told their peers. I received many emails from different demographics saying, ‘thank you for thinking about us.’ It really increased engagement and promoted goodwill within the community.”* Because of limited funding and staffing, Habitat would not have been able to independently design and implement such a tool. The student team effectively expanded the organization’s capacity to

serve a broader population. This reinforces our finding that WCPC plays a crucial role in helping organizations overcome resource limitations and continue serving their communities.

**Thus, this shows that WCPC projects have helped organizations expand their community outreach and access resources** they would not have had without student support. Many sponsors described how student-created tools (such as dashboards, databases, assisted technology, Gantt charts, and models) enabled them to better engage residents and reach audiences they could not reach on their own. In this way, WCPC projects not only address organizational challenges but also enhance the capacity of community partners to serve Worcester more broadly. Our analysis shows that the WCPC's value goes far beyond individual projects. Rather than operating as isolated partnerships, WCPC projects contribute to a broader collaborative ecosystem, where organizations become indirectly connected through overlapping themes, shared initiatives, and sustained engagement.

Across 25 years of work, the Worcester Community Project Center (WCPC) has built a large, durable, and locally embedded network of partners while consistently contributing to three primary areas of community impact: **Environmental & Sustainability, Education & Learning, and Community Development**. Through this long-term engagement, the WCPC has formed a network of more than 160 partners and a focus on the target core impact areas, to track and record the tangible products and practical solutions that students come up with for the City of Worcester. This allowed the Center to track and support the tangible products, practical solutions, and tools that students develop for community partners.

Together, these findings and analyses highlighted that the WCPC's long-term impact lies not only in the number of projects completed but in the lasting relationships, improved communication systems, and strengthened community connections it helped build throughout Worcester and far beyond that. At the same time, our analysis also revealed opportunities to strengthen how this impact is documented, communicated, and sustained over time. Based on these findings, we propose the following recommendations.

## Recommendations

### Revamp the project archive system

In order to better convey WCPC's 25-year history, the Center should strengthen how projects are presented and referenced on its website. While the Gordon Library's Digital



WPI serves as the official archive, the WCPC website functions as the primary public-facing platform and easy access reference point. Currently, the website only includes projects from 2012 to present, even though additional projects dating back to 1992 are available in the Digital WPI archive. Hence, we recommend updating the WCPC website so that it consistently reflects all completed projects, either by uploading summaries directly to the site or by linking each project to its corresponding Digital WPI entry. Thus, WCPC should use our derived themes (Appendix F) as standardized tags for all projects, along with clear project type categories seen in Figure 6 (such as program development, etc), as well as brief project summaries. Because most older projects did not have tags, implementing our standardized themes and project type labels would keep the archive more consistent and improve clarity, making it easier for community members, sponsors, and students to explore WCPC's long-term work. These changes would not replace the library's role as the official archive, but would instead improve accessibility, visibility, and ease of navigation for users who rely primarily on the WCPC website.

## Publish a Public Partners Page

To make partnerships visible beyond individual projects, WCPC should publish a public "Partners" page that lists all organizations, highlights those with the highest number of collaborations, and shows years or counts of engagement. This is not the map we produced; it's an ongoing page WCPC maintains to celebrate the ecosystem, help new sponsors see how they fit, and provide a transparent picture of the Center's network over time. The page can start simple with a searchable directory with a short and be refined annually.

## Create Space for Partners to Connect Through an Annual WCPC Community Gathering

We recommend organizing an annual event that provides an opportunity for project sponsors who worked with the WCPC to share and discuss with each other the stories of community impact and connections with other organizations. The purpose of this event is to provide organizations across Worcester with an opportunity to interact with each other and to network. This event would help the wider community connect and discuss with organizations loosely enough so that participants can have conversations with each

other. This event would ultimately allow participants to reflect and record other organizations they have worked with or interacted with in the past year and strengthen those partnerships overall. In addition to strengthening relationships among community partners, this event would also highlight WPI's role in the community. By showcasing completed projects, sharing impact stories, and recognizing long-term relationships, this gathering would demonstrate how WPI students contribute to local capacity-building efforts and sustained community development. Presentations, visual displays of past projects, or brief sponsor reflections of these different projects could make WPI's involvement more visible to the attendees.

## Embed Community Voice Through Formalized Feedback Processes

We recommend that WCPC establish a formalized feedback process for future projects to ensure that the voices of its partner organizations and community members remain central to its communications. While feedback is often incorporated during the IQP process, we suggest implementing an additional structured feedback opportunity for sponsors after the conclusion of the term. This could include brief post project interviews or surveys that allow sponsors to reflect on their overall experience working with the student team and the WCPC. These tools could gather feedback not only on the clarity of project deliverables and community impact, but also on communication consistency, student engagement, challenges such as travel or scheduling, and whether the project ultimately met the organization's needs. Creating space for sponsors to discuss what worked well and what could be improved would strengthen accountability and long term partnership quality. Implementing this practice would reinforce the university community relationship and serve as a lasting symbol that WCPC listens, adapts, and honors the people it represents.

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# References

- Andreasen, A. R. (2005). *Social Marketing in the 21st Century* (1st ed.). SAGE Publications, Incorporated. <https://doi.org/10.4135/9781483329192>
- Arnstein, S. R. (1969). A ladder of citizen participation. *Journal of the American Planning Association*, 35(4), 216–224. <https://doi.org/10.1080/01944366908977225>
- Benford, R. D., & Snow, D. A. (2000). Framing Processes and Social Movements: An Overview and Assessment. *Annual Review of Sociology*, 26(1), 611–639. <https://doi.org/10.1146/annurev.soc.26.1.611>
- Caprar, D.V., Neville, B.A. (2012). “Norming” and “Conforming”: Integrating Cultural and Institutional Explanations for Sustainability Adoption in Business. *J Bus Ethics* 110, 231–245. <https://doi-org.ezpv7-web-p-u01.wpi.edu/10.1007/s10551-012-1424-1>
- Chazdon, Scott; Emery, Mary; Hansen, Debra; Higgins, Lorie; Sero, Rebecca. (2017). A Field Guide to Ripple Effects Mapping. Retrieved from the University Digital Conservancy, <https://hdl.handle.net/11299/190639>.
- Chumo, I., Kabaria, C., & Mberu, B. (2025). Social mapping of marginalized and vulnerable groups in urban informal settlements: A participatory action research approach. *Frontiers in Sustainable Cities*, 7, 1693073. <https://doi.org/10.3389/frsc.2025.1693073>
- College of Science and Mathematics. (2023). What Makes a HIP a HIP? *Fresnostate.edu*. [https://csm.fresnostate.edu/hips\\_in\\_csm/what-makes-a-hip-a-hip.html](https://csm.fresnostate.edu/hips_in_csm/what-makes-a-hip-a-hip.html)
- Columbia University Mailman School of Public Health. (2026). Content analysis method and examples. *Columbia Public Health*. <https://www.publichealth.columbia.edu/research/population-health-methods/content-analysis>
- De Vreese, R., Leys, M., Fontaine, C. M., & Dendoncker, N. (2016). Social mapping of

- perceived ecosystem services supply – The role of social landscape metrics and social hotspots for integrated ecosystem services assessment, landscape planning and management. *Ecological Indicators*, 66, 517–533.  
<https://doi.org/10.1016/j.ecolind.2016.01.048>
- Diab, P., & Flack, P. (2013). Benefits of community-based education to the community in South African health science facilities. *African Journal of Primary Health Care & Family Medicine*, 5(1), 474. <https://doi.org/10.4102/phcfm.v5i1.474>
- Downs, T. J., Ross, L., Mucciarone, D., Calvache, M.-C., Taylor, O., & Goble, R. (2010). Participatory testing and reporting in an environmental-justice community of Worcester, Massachusetts: a pilot project. *Environmental Health*, 9(1), Article 34.  
<https://doi.org/10.1186/1476-069X-9-34>
- Downs, T. J., Ross, L., Patton, S., Rulnick, S., Sinha, D., Mucciarone, D., Calvache, M., Parmenter, S., Subedi, R., Wysokenski, D., Anderson, E., Dezan, R., Lowe, K., Bowen, J., Tejani, A., Piersanti, K., Taylor, O., & Goble, R. (2009). Complexities of holistic community-based participatory research for a low-income, multi-ethnic population exposed to multiple built-environment stressors in Worcester, Massachusetts. *Environmental Research*, 109(8), 1028–1040.  
<https://doi.org/10.1016/j.envres.2009.08.005>
- Elmes, M., & Loiacono, E. T. (2009). Project-based service-learning for an unscripted world: the WPI IQP experience. *International Journal of Organizational Analysis* (2005), 17(1), 23–39. <https://doi.org/10.1108/19348830910948887>
- Eshuis, J. (2012). *Branding in governance and public management*. Routledge.
- Gertler, P., & World Bank, contributor. (2016). *Impact evaluation in practice* (Second edition.). World Bank.  
<https://www.worldbank.org/en/programs/sief-trust-fund/publication/impact-evaluation-in-practice>
- Greene, M., Stroope, J., & Holston, D. (2022). Adapting Ripple Effect Mapping to a Virtual

- Survey Format. *Journal of Extension*, 60(1). <https://doi.org/10.34068/joe.60.01.10>
- Harris, K., Nobles, J., Ryan, L., Szedlak, C., Taylor, H., Hawkins, R., Cline, A., Smith, E., & Hall, A. (2024). Towards realist-informed ripple effects mapping (RREM): positioning the approach. *BMC Medical Research Methodology*, 24(1), Article 259. <https://doi.org/10.1186/s12874-024-02371-7>
- Hough, L. M. (1984). Development and evaluation of the "accomplishment record" method of selecting and promoting professionals. *Journal of applied psychology*, 69(1), 135. <https://psycnet.apa.org/fulltext/1984-11173-001.pdf>
- Michel, J., & Schneider, K. (2025). Demystifying impact evaluation: an impact evaluation framework. *Frontiers in Epidemiology (Online)*, 5, 1460997. <https://doi.org/10.3389/fepid.2025.1460997>
- Pusca, D., & Northwood, D. (2018). Implementation of high-impact practices in engineering design courses. *World Transactions on Engineering and Technology Education*, 16, 108–114.
- Reilly, S., Langley-Turnbaugh, S., & Neikirk, M. (2023). Building and Broadening Place-Based Community Engagement: A Case Study. *Collaborations (Coral Gables, Fla.)*, 6(1). <https://doi.org/10.33596/coll.108>
- Shrestha, Priyanka, "Urban Permaculture for Climate-Resilient Farming in Worcester, Massachusetts" (2021). *Sustainability and Social Justice*. 254. [https://commons.clarku.edu/idce\\_masters\\_papers/254](https://commons.clarku.edu/idce_masters_papers/254)
- Stephens, J. C., Hernandez, M. E., & Boyle, M. E. (2009). *Learning from university-community partnerships (past and present) for sustainable development* (GPMI Working Papers No. 2009-04). George Perkins Marsh Institute, Clark University.
- Ware, C. (2013). *Information visualization : perception for design* (4th edition.). Morgan Kaufmann is an imprint of Elsevier. <https://doi.org/10.1016/C2009-0-62432-6>
- Weyer, D., Bezerra, J. C., & De Vos, A. (2019). Participatory Mapping in a Developing Country

Context: Lessons from South Africa. *Land*, 8(9), 134.

<https://doi.org/10.3390/land8090134>

# End Notes

<sup>1</sup> Example Project focused on food systems:

Badhwar, A., Sukthankar, A., Kowalski, M., Pu, Y. (2024). *Proposal for Improving the Inventory System of Worcester County Food Bank* [Interactive Qualifying Project, Worcester Polytechnic Institute]. Digital WPI.  
[https://digital.wpi.edu/concern/student\\_works/8k71nn38f?locale=en](https://digital.wpi.edu/concern/student_works/8k71nn38f?locale=en).

<sup>2</sup> Project focused on design implementation:

Bharadwaj, A., Nourse, M., Poncet, S., Vasiliou, M. (2025). *Developing an E-Commerce Inventory System for the Ashland ReStore* [Interactive Qualifying Project, Worcester Polytechnic Institute]. Digital WPI.  
[https://digital.wpi.edu/concern/student\\_works/fq978043z?locale=en](https://digital.wpi.edu/concern/student_works/fq978043z?locale=en).

<sup>3</sup> Project focused on climate planning:

Baker, J., Chmykh, I., Eigenbrodt, C. (2012). *Town of Auburn Climate Action Plan* [Interactive Qualifying Project, Worcester Polytechnic Institute]. Digital WPI.  
[https://digital.wpi.edu/concern/student\\_works/4m90dw029?locale=en](https://digital.wpi.edu/concern/student_works/4m90dw029?locale=en).

<sup>4</sup> Project that creates the weight estimation calculation model:

Gomes, M., Mohareb, J., Larrabee, M. (2023). *Developing A New and Sustainable Weight Estimation Method for Habitat for Humanity ReStore Metro-West/Greater Worcester and Beyond* [Interactive Qualifying Project, Worcester Polytechnic Institute]. Digital WPI. [https://digital.wpi.edu/concern/student\\_works/bg257j62q?locale=en](https://digital.wpi.edu/concern/student_works/bg257j62q?locale=en)

<sup>5</sup> Project that creates the assistive technology database:

Bingel, J., Cornwell, A., DiMilia, M., Meersman, N., Tran, N. (2016). *INCREASING THE ACCESSIBILITY OF ASSISTIVE TECHNOLOGY AT SEVEN HILLS FOUNDATION* [Interactive Qualifying Project, Worcester Polytechnic Institute]. Digital WPI.  
[https://digital.wpi.edu/concern/student\\_works/8s45q9195?locale=en](https://digital.wpi.edu/concern/student_works/8s45q9195?locale=en)

<sup>6</sup> Example project creating a tool:

Goodchild, J., Kiely, K., Reynolds, C. (2025). *Identifying and Implementing a Translation Tool for use in Habitat for Humanity ReStore Locations* [Interactive Qualifying Project,

Worcester Polytechnic Institute]. Digital WPI.

[https://digital.wpi.edu/concern/student\\_works/2v23vz69f?locale=en](https://digital.wpi.edu/concern/student_works/2v23vz69f?locale=en)

<sup>7</sup> Example project creating a model:

Araujo, J. M., DeMalia, M. A., Mandal, S., Piya, C. K. (2007). *Models for Greening Urban Redevelopment Projects: A Case Study of the Blackstone Canal District in Worcester, Massachusetts* [Interactive Qualifying Project, Worcester Polytechnic Institute]. Digital WPI.

[https://digital.wpi.edu/concern/student\\_works/k0698800g?locale=en](https://digital.wpi.edu/concern/student_works/k0698800g?locale=en)

<sup>8</sup> Example project creating a database:

Landrum, E. F., Salazar, E. R., Silva, J. J., Tervola, T. O. (2004). *Brownfield redevelopment database initiative* [Interactive Qualifying Project, Worcester Polytechnic Institute]. Digital WPI.

[https://digital.wpi.edu/concern/student\\_works/ft848q936?locale=en](https://digital.wpi.edu/concern/student_works/ft848q936?locale=en)

<sup>9</sup> Example project improving outreach and communication:

Andres, C., Edmondson, E., Thatcher, T. (2024). *Increasing Engagement, Outreach, and Education for an Online Community for LGBTQ Seniors* [Interactive Qualifying Project, Worcester Polytechnic Institute]. Digital WPI.

[https://digital.wpi.edu/concern/student\\_works/n870zw314?locale=en](https://digital.wpi.edu/concern/student_works/n870zw314?locale=en)



# Appendices

## Appendix A

### Content Analysis

**Purpose:** This appendix explained how we created a structured database from archived WCPC project information using content analysis. The goal was to systematically organize project data into categories that allow us to identify patterns, themes, and relationships across 25 years of WCPC work. This database served as the foundation for a later method for mapping, described in Objective 1. However, the focus was on coding and organizing data that was used later in the project to create different types of maps.

| ID | Project Name                                                                                      | Sponsor                        | Organization                         | Organization Keyword(s)                      | Project Type(s)                          | Project Goal                                                                                                                                                                           | Year | Theme                                                                           |
|----|---------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------|----------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------|
| 1  | Developing an E-Commerce Inventory System for the Ashland ReStore Public Deposited                | Habitat For Humanity           | Park Spirit of Worcester             | Affordable Housing                           | Design Implementation                    | Upon completion of the project, the new system was implemented, and supporting documentation, including training manuals, was created and provided to the ReStore.                     | 2025 | Responsible Consumption and Production                                          |
| 2  | Optimizing Data Collection in Youth Mental Health Education                                       | Charisse Murphy, Julia Freitas | Shine Initiative                     | Mental Health Education                      | Research & Tool Implementation           | Implement a new tool to collect non-clinical mental health data called PowerBI for Shine to develop an instruction manual to ease the organizations' transition to this new framework. | 2025 | Good Health and Well-being<br>Quality Education                                 |
| 3  | Identifying and Implementing a Translation Tool for use in Habitat for Humanity ReStore Locations | Kevin Maguire                  | Habitat for Humanity                 | Language Barrier Translator                  | Enhance Access to Translation Technology | Through building a web-based translation solution, this project offers a low-cost, replicable strategy for nonprofits to enhance access through translation technology.                | 2025 | Reducing Inequality                                                             |
| 4  | Proposal for Improving the Inventory System of Worcester County Food Bank                         | Sarah Walker                   | Worcester County Food Bank           | Food Bank                                    | Improve Inventory Process                | Introduced a barcode checkout system and improved inventory workflows using Agency Express and food bank models.                                                                       | 2024 | Zero Hunger<br>Good Health and Well-being<br>Sustainable Cities and Communities |
| 5  | STEM Education in Worcester: Tracking Community Engagement                                        | Kathy Chen                     | WPI STEM Education Center            | STEM/PreK-12 Education, Community Engagement | Software Development                     | Enhance WPI's system for gathering community engagement data using the software, Collaboratory.                                                                                        | 2024 | Quality Education<br>Industry, Innovation and Infrastructure                    |
| 6  | Converting El Buen Samaritano Food Program Inc. to Digital Client Tracking                        | Maricelis Gonzalez             | El Buen Samaritano Food Program Inc. | Food Insecurity                              | Data Systems Implementation              | Replace paper-based client intake with a barcode-driven digital tracking system and automate reporting to improve food distribution efficiency and funding sustainability              | 2024 | Zero Hunger<br>Reducing Inequality                                              |

#### Materials:

- Microsoft Excel for organizing archived project data into categories.
- WCPC website and Digital WPI

**Procedure for Content Analysis:**

Analyze WCPC Projects -> Divide into Categories in Excel-> Identify Patterns/Themes -> Code

This dataset supported visualization and mapping, but was primarily intended for coding and pattern identification.

Here's the link to the full Excel sheet: [NetworkMap.xlsx](#)

# Appendix B

## Interview Invitation for Project Sponsors

Email Template:

To: **[Sponsor email]**

Cc: [gr-wcpc-c26@wpi.edu](mailto:gr-wcpc-c26@wpi.edu)

Subject: Interview Request for WCPC Project - WPI students

Body:

Hello **[Name]**,

Our names are Malak Haida, Adam Beckman, and Toluwani Akomolafe, and we're part of a student research team working with the Worcester Community Project Center (WCPC). Our project focuses on helping WCPC better document and communicate the long-term community impact of its work across the Greater Worcester area.

Through our review of WCPC's past projects, we noticed that you collaborated on a WCPC-supported project **[Project name]** connected to **[insert relevant theme, example: Food Insecurity]**, particularly through your work with **[Organization Name]**.

We are reaching out to better understand how these collaborations have affected the community beyond the project's completion. Your perspective would help us capture meaningful outcomes, lessons learned, and stories that aren't always reflected in final deliverables or reports.

We were hoping to ask if you would be open to a **brief 20–30 minute interview** to share your insights. We can work around your schedule.

Thank you so much for your time and consideration. We'd truly appreciate the opportunity to learn from your experience.

Best, WCPC team

Contact info: [gr-wcpc-c26@wpi.edu](mailto:gr-wcpc-c26@wpi.edu)

## Appendix C

### List of Sponsoring Organizations We Interviewed

| Organization Name:                                   |
|------------------------------------------------------|
| Seven Hills Foundation                               |
| Coalition for A Health Greater Worcester             |
| Habitat for Humanity                                 |
| WPI STEM Education Center                            |
| Worcester Community Project Center                   |
| Women's Initiative of the United Way                 |
| Mass Audubon                                         |
| Higher Education Consortium of Central Massachusetts |
| Indian Lake Watershed Association                    |
| Refugee Artisans of Worcester                        |

# Appendix D

## Semi-Structured Interview Guide for Sponsors

**Purpose:** To understand the impact that a WCPC project has on a specific theme like Health & Wellness, Environmental Sustainability, and Education, to identify key and specific outcomes, ripple effects, and changes experienced by individuals, organizations, and the greater Worcester community.

Contact Information for WCPC Reference:

General WCPC Contact: [gr-wcpc-c26@wpi.edu](mailto:gr-wcpc-c26@wpi.edu)

Professor Contact: Courtney Kurlanska – [cbkurlanska@wpi.edu](mailto:cbkurlanska@wpi.edu)

### **Procedure:**

This interview will take about 20-30 minutes. I will ask you a series of questions about your experiences and observations. With your permission, I would like to audio-record this conversation to ensure I accurately capture your thoughts. The recording will be stored securely.

### **Confidentiality & Voluntariness:**

Your participation is completely voluntary. You may skip any question or stop the interview at any time. Your responses will be kept confidential. In any reports, we will use a pseudonym or refer to you generally as a 'community leader' unless you give us specific permission to use your name and title.

**Questions?** Do you have any questions for me before we begin?

**Consent:** Do I have your permission to begin the audio recording? And do you consent to participate in this interview?"

### **Interview Questions:**

### **Organization Questions:**

1. Can you tell me about your role in the organization (name)?
2. Did you learn anything from this organization about the issue this project was working on?
3. Who else was impacted because of that change?
4. Did it change access to opportunities?
5. What challenges do you see the program addressing, and how effectively are they managed?
6. Did working with WPI on these projects connect you with other organizations?

**Project Questions:**

7. Did this project have a long-lasting impact? Why or why not?
8. Did this project make a difference?
9. What improvements or changes have you seen because of this project?
10. Did anything change outside the original goals of the project?
11. Can you provide your perspective on working with student researchers on this project?
12. This project was meant to help xyz. How did this xyz impact xyz?
13. Has the project inspired new ideas?
14. Did this project influence relationships, skills, resources, and community spaces?
15. Which parts of the program have been most meaningful to you, and why?
16. Were there any outcomes you did not expect, and how did they occur?
17. Are there any additional effects of the program that are not immediately obvious?
18. What personal challenges did you face with this whole experience?

# Appendix E

## Object-Testing Response

### **Purpose:**

To gather feedback from community members, organizations, and WCPC representatives on the specific materials, findings, and draft presentations we are developing. This feedback will help ensure that our final storytelling media accurately reflects participants' experiences, is accessible, meaningful, and highlights the aspects of WCPC's work they find most important.

### **Procedure:**

We will hold feedback sessions (around 30-60 minutes) where participants can review draft materials, provide feedback on clarity, accessibility, and emphasis, and discuss which information they feel should be highlighted. A note-taker will record major themes and suggestions, and the sessions will also be audio-recorded.

### **Participants:**

Community members, organizational representatives, and WCPC staff who are familiar with WCPC projects or have experience participating in them.

### **Guide:**

- Short introduction/consent script:
  - Thank you for participating in our interview today. We are working on a project with the WCPC to better understand how we should share and promote the data we are collecting about the WCPC's impact over the last twenty-five years.
  - Your participation is completely voluntary. You may skip any question or stop at any time. We will not record your name, and all responses will remain anonymous. We will be taking notes, but no identifying information will be included. Do you have any questions before we start?

Show participant(s) our deliverable website.

- Guiding questions:
  - What is your first impression of this material? Does it capture your attention?
  - What storytelling format feels most engaging to you?

- Looking at the draft materials, what specific elements make the information about community impact easy to understand?
- What, if anything, in these materials and images makes the information about impact confusing or hard to follow?
- Does this website reflect the program's work accurately?
- What style should WCPC use, and how does the style of these draft materials match that ideal?
- Are there any elements that feel unnecessary or distracting?

Scenario: We will now show you our website. Please share your immediate reactions, what works well, and what could be improved.



# Appendix F

## Derived Themes Table

**Purpose:** This appendix represents the consolidated thematic categories developed from the coding of the qualitative data collected throughout the WCPC projects and our content analysis. The Derived Themes Table groups related ideas into high-level themes, provides concise descriptions for each theme, and includes frequency counts representing the number of times these ideas appeared across data sources. Higher counts indicate topics with greater repeated emphasis during the engagement process.

| Themes                                   | Description                                                                   | Count |
|------------------------------------------|-------------------------------------------------------------------------------|-------|
| Accessibility                            | Assistive technology, accessibility services                                  | 13    |
| City Planning                            | Zoning, regional planning                                                     | 11    |
| Health & Wellness                        | Mental health, caregiving, public health, well-being                          | 18    |
| Education & Learning                     | Academic Institutions, STEAM, literacy, curriculum                            | 38    |
| Environment & Sustainability             | Climate, renewable energy, conservation, water, waste                         | 50    |
| Social Equity & Inclusion                | Refugees, immigrants, gender equity, LGBTQ+, racial equity                    | 12    |
| Housing & Infrastructure                 | affordable housing, shelter, infrastructure                                   | 16    |
| Organizational Development & Non-profits | Marketing, nonprofit management, entrepreneurship, organization effectiveness | 16    |
| Food Insecurity                          | Food banks, food hubs                                                         | 12    |
| Economic Growth                          | Local Economy                                                                 | 9     |
| Culture & Arts                           | Museums, cultural exhibits, art, historic preservation                        | 26    |
| Community Development                    | Civic engagement, volunteering, neighborhood engagement                       | 34    |
| Urban Planning & Mobility                | Transportation, public spaces, parks, urban planning                          | 9     |

## Organization Word Cloud

**Purpose:** The image below is a word cloud containing the names of all of the organizations the WCPC has collaborated with during the WCPC's history (168). The size of each word containing the organization name is dependent on how many times that organization has collaborated with the WCPC on a project.

